

Safety data sheet

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BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 05.04.2023

Version: 3.0

Product: **Kollidon® CL-F**

(ID no. 30274401/SDS_GEN_00/EN)

Date of print 20.05.2026

1. Identification

Product identifier

Kollidon® CL-F

Chemical name: 2-Pyrrolidinone, 1-ethenyl-, homopolymer

CAS Number: 9003-39-8

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: pharmaceutical excipient

Details of the supplier of the safety data sheet

Company:

BASF SE

67056 Ludwigshafen

GERMANY

Operating Division Nutrition and Health

Telephone: +49 621 60-48434

E-mail address: EN-global-safety-data@basf.com

Emergency telephone number

International emergency number:

Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

No need for classification according to GHS criteria for this product.

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Label elements

Globally Harmonized System (GHS)

The product does not require a hazard warning label in accordance with GHS criteria.

Other hazards

According to UN GHS criteria

The product is under certain conditions capable of dust explosion.

3. Composition/Information on Ingredients

Substances

Chemical nature

polyvinylpyrrolidone, crosslinked

Hazardous ingredients (GHS)

According to UN GHS criteria

| No particular hazards known.

Mixtures

Not applicable

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air.

On skin contact:

Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:

Rinse mouth and then drink 200-300 ml of water.

Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

5. Fire-Fighting Measures**Extinguishing media**

Suitable extinguishing media:

| foam, dry powder, water spray

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:

| Avoid whirling up the material/product because of the danger of dust explosion.

Special hazards arising from the substance or mixture

carbon oxides, nitrogen oxides

The substances/groups of substances mentioned can be released in case of fire. Evolution of fumes/fog.

Advice for fire-fighters

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

| Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.
| Cool endangered containers with water-spray.

6. Accidental Release Measures

| Avoid the formation and build-up of dust - danger of dust explosion. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition. Avoid dispersal of dust in the air (e.g. by clearing dusty surfaces with compressed air).

Personal precautions, protective equipment and emergency procedures

| Use personal protective clothing. Information regarding personal protective measures, see section 8.
| Avoid dust formation. Avoid contact with the skin, eyes and clothing.

Environmental precautions

Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

| For small amounts: Contain with dust binding material and dispose of.
| For large amounts: Sweep/shovel up.
| Dispose of absorbed material in accordance with regulations. Avoid raising dust.

7. Handling and Storage

Precautions for safe handling

Avoid dust formation. Provide exhaust ventilation if dust is formed.

Protection against fire and explosion:

The product is capable of dust explosion. Avoid dust formation. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy. Use explosion-proof apparatus and fittings.

Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container tightly closed and dry.

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

9003-39-8: 2-Pyrrolidinone, 1-ethenyl-, homopolymer, cross-linked

Exposure controls

Personal protective equipment

Respiratory protection:

Breathing protection if breathable aerosols/dust are formed. Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1 or FFP1)

Hand protection:

Chemical resistant protective gloves (EN ISO 374-1)

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. Avoid contact with the skin, eyes and clothing. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	powder
Colour:	white
Odour:	almost odourless
pH value:	
	substance/mixture is non-soluble (in water)
onset of melting:	130 °C (1.013 hPa)
Boiling point:	not applicable
Flash point:	not applicable, the product is a solid
Evaporation rate:	
Flammability:	The product is a non-volatile solid. not highly flammable (UN Test N.1 (ready combustible solids))
Lower explosion limit:	
	For solids not relevant for classification and labelling.
Upper explosion limit:	
	For solids not relevant for classification and labelling.
Vapour pressure:	
	negligible
Density:	
	No information is available for the absolute density. Instead the bulk density was determined as a more relevant value.
Relative vapour density (air):	
	The product is a non-volatile solid.
Solubility in water:	insoluble
Solubility (qualitative) solvent(s):	organic solvents
	insoluble
Partitioning coefficient n-octanol/water (log Kow):	
	not determined
Thermal decomposition:	>= 150 °C (DSC (DIN 51007))
	No decomposition if stored and handled as prescribed/indicated.
Viscosity, kinematic:	
	not applicable, the product is a solid
Explosion hazard:	Product is not explosive, however a dust explosion could result from an air / dust mixture.
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.

Other information

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Burning rate:	200 mm, 601 s The material doesn't meet the criteria specified in paragraph 33.2.4.4 of UN manual of tests and criteria. An ignition of the test substance does not lead to a propagation by burning with flame or smouldering.	(UN Test N.1 (ready combustible solids))
Self heating ability:	It is a substance capable of spontaneous heating according to UN transport regulations class 4.2. Based on test results packaging < 450 l are exempted from the classification.	(UN Test N.4 (self heating substances))
SADT:	Not a substance/mixture liable to self-decomposition according to GHS.	
Minimum ignition energy:	10 - 30 mJ (1.013 hPa, 20 °C) Inductivity: 0,1 mH The product is capable of dust explosion.	(VDI 2263, sheet 1, 2.5)
Bulk density:	approx. 210 - 230 kg/m ³ (20 °C)	

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: Corrosive effects to metal are not anticipated.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

Dust can form an explosive mixture with air.

Conditions to avoid

See SDS section 7 - Handling and storage. Avoid dust formation. Avoid electro-static charge.

Incompatible materials

Substances to avoid:

None known during use and storage if used according to instructions.

Hazardous decomposition products

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

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Acute toxicity

Assessment of acute toxicity:

Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation.

Experimental/calculated data:

LD50 rat (oral): > 2.000 mg/kg (BASF-Test)

LC50 rat (by inhalation): > 5,2 mg/l 4 h (OECD Guideline 403)

Irritation

Assessment of irritating effects:

Not irritating to the skin. Not irritating to the eyes.

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (Draize test)

Serious eye damage/irritation rabbit: non-irritant (Draize test)

Respiratory/Skin sensitization

Assessment of sensitization:

No data available.

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in studies with mammals.

Carcinogenicity

Assessment of carcinogenicity:

In long-term animal studies in which the substance was given in high doses by feed, a carcinogenic effect was not observed.

Reproductive toxicity

Assessment of reproduction toxicity:

No data available.

Developmental toxicity

Assessment of teratogenicity:

No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Specific target organ toxicity (single exposure)

Remarks: No data available.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

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No data available.

Experimental/calculated data:
not determined

Aspiration hazard

| not applicable

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from products of a similar structure and composition.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:

LC50 (96 h) > 10.000 mg/l, *Leuciscus idus* (DIN 38412 Part 15, static)

Microorganisms/Effect on activated sludge:

EC20 (0,5 h) > 1.995 mg/l, activated sludge, industrial (OECD Guideline 209, aerobic)

Assessment of terrestrial toxicity:

No data available concerning terrestrial toxicity.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Poorly eliminated from water.

Elimination information:

< 10 % DOC reduction (15 d) (OECD Guideline 302 B) (aerobic, activated sludge, industrial) Poorly eliminated from water.

Bioaccumulative potential

Bioaccumulation potential:

Based on its structural properties, the polymer is not biologically available. Accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:

Volatility: No data available.

Adsorption in soil: No data available.

Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Add. remarks environm. fate & pathway:

The product has not been tested. The statements on environmental fate and pathway have been derived from products of a similar structure and composition.

Other ecotoxicological advice:

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

13. Disposal Considerations

Waste treatment methods

Observe national and local legal requirements.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Land transport

ADR

UN number or ID number: UN3088

UN proper shipping name: SELF-HEATING SOLID, ORGANIC, N.O.S.
(POLYVINYL POLYPYRROLIDONE)

Transport hazard class(es): 4.2

Packing group: III

Environmental hazards: no

Special precautions for user: Tunnel code: E

Not dangerous goods of class 4.2 in packages up to 450 litres capacity.

RID

UN number or ID number: UN3088

UN proper shipping name: SELF-HEATING SOLID, ORGANIC, N.O.S.
(POLYVINYL POLYPYRROLIDONE)

Transport hazard class(es): 4.2

Packing group: III

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Environmental hazards:	no
Special precautions for user:	Not dangerous goods of class 4.2 in packages up to 450 litres capacity.

Inland waterway transport

ADN

UN number or ID number:	UN3088
UN proper shipping name:	SELF-HEATING SOLID, ORGANIC, N.O.S. (POLYVINYL POLYPYRROLIDONE)
Transport hazard class(es):	4.2
Packing group:	III
Environmental hazards:	no
Special precautions for user:	Not dangerous goods of class 4.2 in packages up to 450 litres capacity.

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

UN number or ID number:	UN 3088
UN proper shipping name:	SELF-HEATING SOLID, ORGANIC, N.O.S. (POLYVINYL POLYPYRROLIDONE)
Transport hazard class(es):	4.2
Packing group:	III
Environmental hazards:	no
	Marine pollutant: NO
Special precautions for user:	EmS: F-A; S-J Not dangerous goods of class 4.2 in packages up to 450 litres capacity.

Air transport

IATA/ICAO

UN number or ID number:	UN 3088
UN proper shipping name:	SELF-HEATING SOLID, ORGANIC, N.O.S. (POLYVINYL POLYPYRROLIDONE)
Transport hazard class(es):	4.2
Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for	Not dangerous goods of class 4.2 in packages up to 450 litres

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user: capacity.

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

Further information

Not dangerous goods of class 4.2 in packages up to 450 litres capacity.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

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